NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY

ATTA-UR-RAHMAN SCHOOL OF SCIENCE AND TECHNOLOGY

PROJECT REPORT

INDIGENOUS DEVELOPMENT OF DNA TAGGING SYSTEM FOR INDUSTRIES

May, 2026

Project Approvals

This written technical brief documents the system architecture, operational workflow, mathematical calculations, and empirical qPCR verification of the **DNA Base Tag and Trace System**.

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Chapter 1: Project Background & Objectives

1.1 Introduction and Strategic Context

Product counterfeiting, illicit trade, and unauthorized tampering pose significant risks to industrial supply chains, consumer safety, and commercial brand integrity. Critical sectors—including pharmaceuticals, agrochemicals, fuels, and consumer packaged goods—frequently suffer from fraudulent reproduction and substitution.

Conventional security mechanisms, such as overt holograms, standard serial numbers, and 2D barcodes, authenticate only the outer packaging surface. Because these optical markers can be physically copied, photographed, or transferred, they fail to provide definitive, tamper-proof security.

To address these challenges, the National University of Sciences and Technology (NUST) initiated the **Indigenous DNA Tagging System for Industries**. By utilizing trace quantities of a unique, synthetic DNA construct as a covert molecular marker, products can be authenticated with forensic certainty using polymerase chain reaction (PCR) and real-time quantitative PCR (qPCR) detection.

1.2 Conventional Anti-Counterfeiting vs. DNA Tagging

Table 1.1 summarizes the primary distinctions between conventional security markers and covert synthetic DNA taggants.

Table 1.1. Comparison: Conventional Surface Security vs. Synthetic DNA Tagging

Parameter	Conventional Security (Barcodes, Holograms)	Synthetic DNA Tagging System
Visibility	Overt / Visible surface marker (easily replicated or photographed).	**Covert**: Invisible micro-trace applied at parts-per-billion or femtogram levels.
Security Level	Low to moderate (vulnerable to digital cloning and physical detachment).	**Unforgeable**: Defined synthetic nucleotide sequence readable only with specific primers and probes.
Product Integration	Surface label or packaging sticker only.	Integrated directly onto packaging surfaces, QR codes, or liquid formulations.
Verification	Visual inspection or optical scanner.	**Forensic**: Confirmatory PCR amplification and real-time qPCR probe deconvolution.

1.3 Specific Objectives of the Project

The primary objective of this project phase is to establish a functional, reproducible laboratory workflow for the production, bulk formulation, product application, and forensic detection of an indigenous DNA taggant:

1. **Construct Recovery and Bacterial Propagation**: To culture and recover the ordered 200 bp synthetic DNA taggant construct supplied in transformed *Escherichia coli* (Stab Top10) in pUC57 plasmid vector.
2. **PCR Amplification and Gel Verification**: To optimize enzymatic amplification conditions using universal primer pairs and verify construct size on 2.0% agarose gel electrophoresis.
3. **Serial Dilution Optimization**: To evaluate serial 1:10 dilutions (D_1 to D_4) of the amplicon, determine the optimal working dilution, and resolve template-dependent qPCR inhibition.
4. **1.0-Liter Bulk Carrier Formulation**: To prepare a standardized 1.0-Liter formulation of stabilized Tris-EDTA (TE) buffer containing the micro-dosed DNA taggant suitable for industrial application.
5. **Spatial Homogeneity Assessment**: To verify uniform distribution and stability of the DNA taggant across multiple fractions (F1 to F7) of the 1.0-Liter bulk preparation using real-time qPCR.
6. **Sequence-Specific Probe Optimization**: To evaluate four distinct fluorophore probes (Probes A, B, C, and D) in individual and 4-plex multiplex real-time qPCR formats.
7. **Commercial Product Application**: To apply the formulated DNA taggant onto industrial packaging materials (syrup bottles, cartons, blister foils) associated with 2D QR codes, as well as liquid formulations, and demonstrate forensic recovery via surface swabbing.

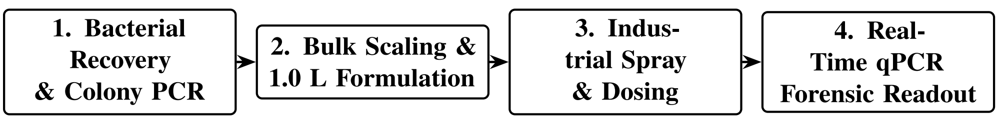


Figure 1.1. Four-stage operational workflow of the Indigenous DNA Tagging System.

Chapter 2: Materials & Taggant Synthesis

2.1 DNA Taggant Construct & Bacterial Host

The molecular marker developed for this project is a unique, synthetic 200 bp double-stranded DNA sequence designed specifically for covert industrial authentication. The construct was supplied cloned into the pUC57 plasmid vector with ampicillin selection (Lot No. BJ0335352-1, miniprep yield ~ 4 μg) and hosted in transformed *Escherichia coli* (Stab Top10).

(a) Cloned Taggant (pUC57) **(b)** Forward Primer Stock **(c)** Reverse Primer Stock

Figure 2.1. Physical synthesis materials: (a) Cloned 200 bp synthetic DNA taggant template in pUC57 plasmid vector (Lot No. BJ0335352-1, ~ 4 μg miniprep), (b) Universal Forward Primer stock (T_m = 58.5°C, reconstitute in 194.4 μL), and (c) Universal Reverse Primer stock (reconstitute in 180.4 μL).

2.2 Bacterial Culture & Colony Heat-Lysis

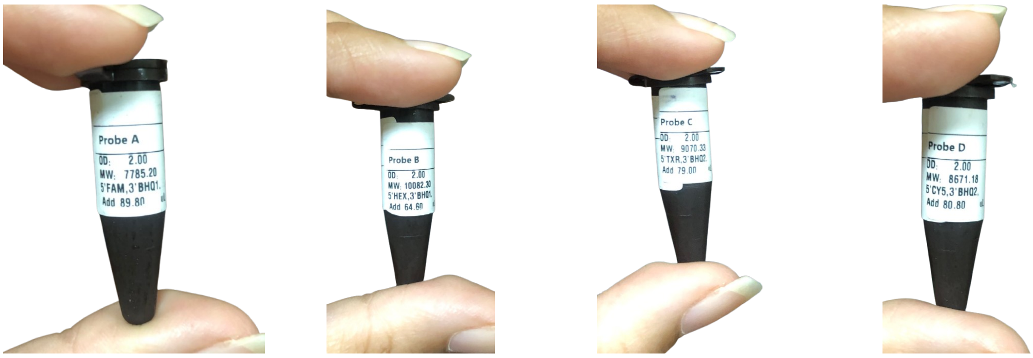
1. **Cultivation**: Transformed Stab Top10 *E. coli* cells were cultured in 10 mL Luria-Bertani (LB) broth supplemented with ampicillin (100 μg/mL) in a shaking incubator at 37°C for 16 hours. Solid cultures were maintained on ampicillin-selective LB agar plates.

2. **Rapid Heat-Lysis**: Rather than utilizing commercial plasmid extraction kits, individual bacterial colonies were suspended in 100 μL of sterile TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0). Heat lysis was executed in a thermal cycler at 95°C for 10 minutes. Following a brief centrifugation, the supernatant was used directly as template for PCR amplification.

2.3 Universal Primers & Sequence-Specific Probes

Enzymatic amplification and real-time quantification utilize standardized universal primers and four sequence-specific fluorophore-labeled hydrolysis probes:

- **Universal Forward Primer**: T_m = 58.5°C (reconstituted in 194.4 μL nuclease-free water for 100 μM stock).
- **Universal Reverse Primer**: Reconstituted in 180.4 μL nuclease-free water for 100 μM stock.
- **Sequence-Specific Optical Probes (A, B, C, D)**: Four distinct probes targeting internal regions of the 200 bp taggant construct (Figure 2.2 and Table 2.1).



(a) Probe A (FAM) **(b)** Probe B (HEX) **(c)** Probe C (TxRed) **(d)** Probe D (Cy5)

Figure 2.2. Physical synthesized fluorophore probe stocks: (a) Probe A (5′-FAM, 3′-BHQ1), (b) Probe B (5′-HEX, 3′-BHQ1), (c) Probe C (5′-Texas Red, 3′-BHQ2), and (d) Probe D (5′-Cy5, 3′-BHQ2).

Table 2.1. Physical Specifications of Synthesized 4-Channel Multiplex Probes

Probe	5’ Fluorophore	3’ Quencher	OD	MW (g/mol)	Reconstitution
Probe A	FAM	BHQ1	2.00	7785.20	Add 89.80 μL
Probe B	HEX	BHQ1	2.00	10082.30	Add 64.60 μL
Probe C	Texas Red (TXR)	BHQ2	2.00	9070.33	Add 79.00 μL
Probe D	Cy5	BHQ2	2.00	8671.18	Add 80.80 μL

2.4 Endpoint PCR Synthesis Formulation & Thermal Cycling

To mass-produce the 200 bp synthetic taggant construct from the bacterial lysate, standard endpoint PCR amplification was conducted in a total volume of 25.0 μL (Figure 2.3, Table 2.2, and Table 2.3).

Thermal Cycling

Step	Temperature	Time	Cycles
Initial Denaturation	95 °C	5 min	1
Denaturation	95 °C	30 sec	
Annealing	55 °C	30 sec	35*
Extension	72 °C	45 sec	
Final Extension	72 °C	7 min	1
Hold	4 °C	∞	1

PCR Recipe

Component	Reaction 1	Reaction 2
10X Buffer	2.5 µL	2.5 µL
$MgCl_2$	1.5 µL	1.5 µL
Forward Primer	1.0 µL	1.0 µL
Reverse Primer	1.0 µL	1.0 µL
Taq DNA Polymerase	1.0 µL	1.0 µL
Template DNA	1.0 µL	2.0 µL
Distilled Water	16.0 µL	15.0 µL
Total Volume	25 µL	25 µL

Figure 2.3. Laboratory master recipe and thermal cycling schedule executed for endpoint PCR amplification.

Table 2.2. Endpoint PCR Synthesis Formulation (25.0 μL Reaction Volume)

Component	Reaction 1	Reaction 2
10× PCR Buffer	2.5 μL	2.5 μL
$MgCl_2$	1.5 μL	1.5 μL
Forward Primer	1.0 μL	1.0 μL
Reverse Primer	1.0 μL	1.0 μL
Taq DNA Polymerase	1.0 μL	1.0 μL
Template DNA	1.0 μL	2.0 μL
Distilled Water	16.0 μL	15.0 μL
Total Volume	**25.0 μL**	**25.0 μL**

Table 2.3. Endpoint Thermal Cycler Parameter Schedule

Step	Temperature	Time	Cycles
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	30 sec	
Annealing	55°C	30 sec	35 cycles*
Extension	72°C	45 sec	
Final Extension	72°C	7 min	1
Hold	4°C	∞	1

*35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).

Chapter 3: Formulation & Industrial Application

3.1 1.0-Liter Bulk Carrier Formulation in TE Buffer

To enable industrial-scale application across commercial products, a standardized 1.0-Liter carrier formulation was prepared using sterile Tris-EDTA (TE) buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0). TE buffer maintains long-term DNA integrity by chelating divalent metal ions (such as Mg^{2+}) that serve as co-factors for environmental DNases.

The master formulation recipe executed in the laboratory is illustrated in Figure 3.1 and summarized in Table 3.1.

1 Liter of TE Buffer containing taggant is a mixture of

1. 10ml of 1M Tris-HCl
2. 2ml of 0.5M EDTA (pH = 8)
3. 988ml of distilled water
4. 100 ul of PCR product (4-5 tubes of PCR products)

Figure 3.1. Laboratory master batch recipe for the 1.0-Liter stabilized TE buffer taggant carrier formulation.

Table 3.1. 1.0-Liter Bulk TE Buffer Carrier and Taggant Dosing Formulation

Component Reagent	Dispensed Volume	Final Concentration
1.0 M Tris-HCl (pH = 8.0)	10.0 mL	10.0 mM
0.5 M EDTA (pH = 8.0)	2.0 mL	1.0 mM
Nuclease-Free Distilled Water	988.0 mL	Solvent Matrix
Amplified DNA Taggant (PCR Product)	100.0 μL (4–5 tubes)	Micro-Dosed Molecular Marker
Total Master Formulation Volume	**1,000.1 mL** ($\approx$ 1.0 **Liter**)	Stabilized Industrial Spray

3.1.1 Sterilization & Mixing Procedure

1. **Buffer Preparation**: 10.0 mL of 1 M Tris-HCl and 2.0 mL of 0.5 M EDTA were measured and added to 900 mL of distilled water. The solution was adjusted to pH = 8.0 and brought to a total volume of 1000 mL.

2. **Autoclaving**: The carrier buffer was sterilized in an autoclave at 121°C for 20 minutes to eliminate any biological contaminants and allowed to cool to ambient temperature.

3. **Taggant Dosing**: 100.0 μL of amplified 200 bp DNA taggant product (pooled from 4–5 tubes of endpoint PCR reactions) was introduced into the 1.0-Liter container.

4. **Homogenization**: The container was agitated on a laboratory blood mixer for continuous rotational blending, achieving uniform spatial distribution of the DNA molecules throughout the 1.0-Liter volume.

3.2 Application to Commercial Products & Packaging

The formulated 1.0-Liter DNA taggant spray was tested across diverse commercial packaging surfaces and liquid formulations:

- **Pharmaceutical Packaging & Blister Foils**: Applied directly to commercial pharmaceutical packaging materials, including Macter syrup bottles, Bismol cartons, and tablet blister foils.

- **2D QR Code Integration**: The taggant was applied over printed 2D DataMatrix and QR code zones on product labels, associating the physical bio-molecular tag with digital identification strings.

- **Direct Liquid Micro-Dosing**: Trace quantities of the taggant solution were introduced into liquid consumer formulations (*Gelsemium 30, Surkhana*) at parts-per-billion concentrations, preserving chemical clarity, physical stability, and product potency.

3.3 Field Sampling & Forensic Swabbing Protocol

To authenticate tagged products in the field without destroying packaging:

1. A sterile cotton swab moistened with sterile TE buffer is firmly rubbed across the marked QR code or packaging surface.
2. The swab head is immersed in 25.0–50.0 μL of sterile nuclease-free water or TE buffer in a microcentrifuge tube.
3. The eluted liquid is used directly as template (1.0–2.0 μL) for confirmatory real-time qPCR forensic deconvolution.

Chapter 4: Experimental Results & Verification

4.1 Agarose Gel Electrophoresis Verification

To verify the molecular size and fidelity of the synthesized taggant construct, amplified products from gradient PCR (54°C–59°C) were evaluated on a 2.0% agarose gel stained with ethidium bromide (Figure 4.1).

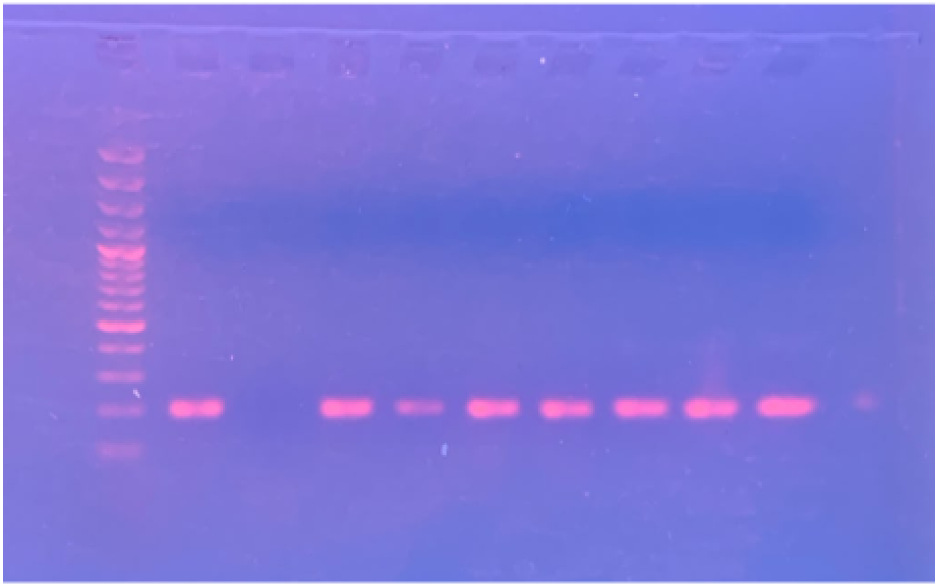


Figure 4.1. Empirical 2.0% agarose gel electrophoresis photograph under UV transillumination showing the DNA ladder on the left (Lane M) and the discrete, high-yield 200 bp amplified DNA taggant bands across gradient annealing lanes.

The electrophoretic run confirmed:

- A single, sharp amplicon band migrated precisely at the 200 bp reference ladder rung.
- Optimal amplification yield was achieved at annealing temperatures between 55°C and 57°C.
- Total absence of non-specific secondary amplicons, primer-dimer complexes, or chimeric artifacts, verifying the high specificity of the universal primer pair.

4.2 Serial Dilution Analysis & Template Dynamics

Real-time qPCR exhibits high analytical sensitivity. In early laboratory testing, undiluted, concentrated PCR product showed partial reaction inhibition. To resolve this and establish the optimal working concentration, serial 1:10 dilutions (D_1 to D_4) were evaluated:

- The concentrated template caused partial kinetic suppression during real-time amplification.
- Serial dilution systematically relieved inhibition, with the 1:10,000 dilution (D_4) producing sharp, reproducible quantification cycles ($C_q \approx 4.78$).
- Consequently, the 1:10,000 dilution was established as the standardized working concentration for bulk carrier formulation.

4.3 1.0-Liter Bulk Homogeneity & Product Recovery

To assess whether the DNA taggant remains uniformly dispersed throughout the 1.0-Liter TE buffer carrier without settling or precipitation, multiple independent fractions (F1 to F7) and product recovery swabbing samples (PR1–PR2) were evaluated via real-time qPCR (Table 4.1).

Table 4.1. qPCR Assessment of Spatial Homogeneity Across 1.0-Liter Bulk Reagent

Sample Fraction	Template Volume	Observed C_q	Homogeneity Status
F1	1.0 μL	**4.21**	Positive (Uniform Dispersion)
F2	1.0 μL	**5.03**	Positive (Uniform Dispersion)
F3	1.0 μL	**5.75**	Positive (Uniform Dispersion)
F4	1.0 μL	**5.34**	Positive (Uniform Dispersion)
F5 (Replicates R1/R2)	1.0–2.0 μL	**5.03 / 6.01**	Positive (Uniform Dispersion)
F6 (Replicates R1/R2)	1.0–2.0 μL	**5.75 / 6.59**	Positive (Uniform Dispersion)
F7 (Replicates R1/R2)	1.0–2.0 μL	**5.34 / 5.36**	Positive (Uniform Dispersion)
PR (Replicates R1/R2)	1.0–2.0 μL	**3.99 / 5.23**	Positive (Product Recovery)

The consistent cycle threshold values (C_q = 3.99–6.59) across all sampled fractions and swabbed packaging surfaces confirm uniform liquid dispersion and stable recovery of the DNA taggant.

4.4 Real-Time qPCR Formulation & Cycling Parameters

Forensic detection was standardized to a 20.0 μL reaction volume utilizing commercial real-time master mix, universal primers, and sequence-specific probes (Figure 4.2,

Table 4.2, and Table 4.3).

PCR Recipe

Component	Reaction 1	Reaction 2
Master Mix	10.0 µL	10.0 µL
Forward Primer	0.6 µL	0.6 µL
Reverse Primer	0.6 µL	0.6 µL
Template DNA	1.0 µL	2.0 µL
ddH_2O	7.8 µL	6.8 µL
Total Volume	20 µL	20 µL

Thermal Cycling

Step	Temperature	Time	Cycles
Initial Denaturation	95 °C	5 min	1
Denaturation	95 °C	30 sec	
Annealing	55 °C	30 sec	35*
Extension	72 °C	45 sec	
Final Extension	72 °C	7 min	1
Hold	4 °C	∞	1

**35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).*

Figure 4.2. Laboratory master recipe and thermal cycler parameter schedule for real-time qPCR verification.

Table 4.2. Real-Time qPCR Reaction Formulation (20.0 μL Total Volume)

Component	Reaction 1	Reaction 2
Master Mix (2×)	10.0 μL	10.0 μL
Forward Primer	0.6 μL	0.6 μL
Reverse Primer	0.6 μL	0.6 μL
Template DNA	1.0 μL	2.0 μL
Distilled Water (ddH_2O)	7.8 μL	6.8 μL
Total Volume	**20.0 μL**	**20.0 μL**

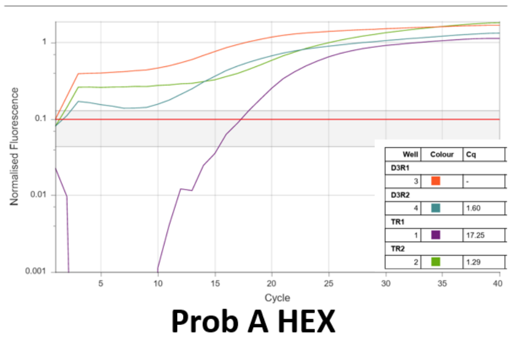
Table 4.3. Real-Time Thermal Cycler Parameter Schedule

Step	Temperature	Time	Cycles
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	30 sec	
Annealing	55°C	30 sec	35 cycles*
Extension	72°C	45 sec	
Final Extension	72°C	7 min	1
Hold	4°C	∞	1

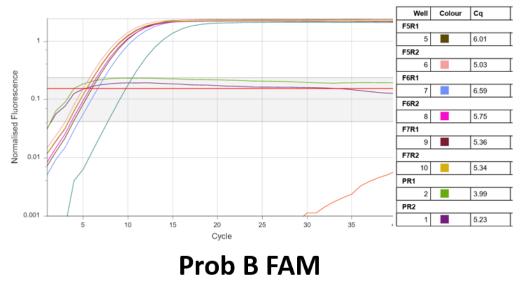
*35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).

4.5 Real-Time qPCR Probe Verification Results

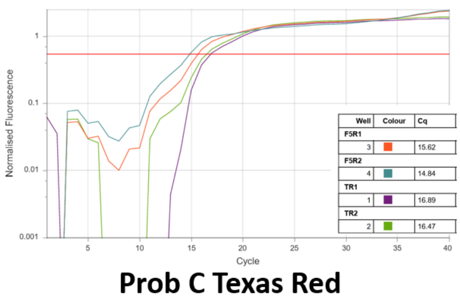
Real-time qPCR amplification was evaluated across all four individual fluorophore probe channels, followed by a simultaneous 4-plex multiplex reaction. Figure 4.3 displays the amplification curves and quantification cycle (C_q) data generated directly by the real-time thermal cycler.



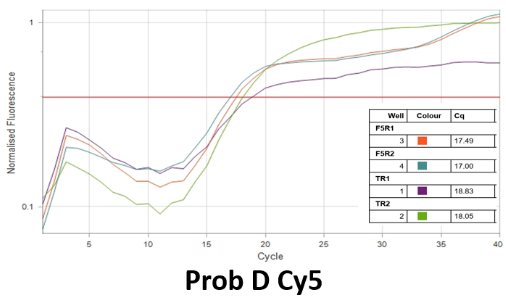
(a) Probe A (HEX Channel)



(b) Probe B (FAM Channel)



(c) Probe C (Texas Red Channel)



(d) Probe D (Cy5 Channel)

Figure 4.3. Empirical real-time qPCR amplification curves and quantification cycle (C_q) data tables captured directly from the thermal cycler across all four optical channels: Probe A (HEX), Probe B (FAM), Probe C (Texas Red), and Probe D (Cy5).

Table 4.4 summarizes the quantification cycle ranges and analytical outcomes across all individual channels and the multiplex format.

Table 4.4. Summary of Empirical Real-Time qPCR Optical Results

Assay Format	Optical Channel	Ex / Em (λ)	Observed C_q Range	Status
Probe A	HEX	535/556 nm	**1.29 – 17.25**	Positive
Probe B	FAM	495/520 nm	**3.99 – 6.59**	Positive
Probe C	Texas Red	585/610 nm	**14.84 – 16.89**	Positive
Probe D	Cy5	650/670 nm	**17.00 – 18.83**	Positive
4-Plex Multiplex	4 Channels	Multi-Channel	**11.95 – 15.49**	Positive

The positive amplification across all four probe channels in both individual and multiplex configurations confirms that the synthetic DNA taggant can be authenticated with complete molecular specificity.

Chapter 5: Discussion & Conclusion

5.1 Discussion of Experimental Findings

The laboratory results documented in this report demonstrate the successful implementation of an end-to-end indigenous DNA taggant system for industrial product authentication:

1. **Construct Recovery & Amplification Fidelity**: The synthetic 200 bp DNA taggant construct was successfully recovered from transformed *Escherichia coli* (Stab Top10) in pUC57 plasmid vector using a rapid colony heat-lysis procedure. Conventional PCR and 2.0% agarose gel electrophoresis confirmed the presence of a single, sharp 200 bp band without secondary amplicons or primer-dimer artifacts, verifying high enzymatic specificity.
2. **Resolution of Concentration Effects**: Testing of the concentrated PCR template revealed partial reaction inhibition during real-time qPCR. Serial 1:10 dilution analysis resolved this inhibition, identifying the 1:10,000 dilution (D_4) as the optimal working concentration ($C_q \approx 4.78$). This finding established a scientific basis for micro-dosing the taggant at trace levels.
3. **Large-Volume Carrier Homogeneity**: The preparation of a 1.0-Liter bulk carrier in stabilized TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0) containing 100 μL of pooled amplicon yielded consistent real-time qPCR detection across all sampled fractions (F1 to F7, C_q = 4.21–5.75). This demonstrates that rotational blending using a laboratory blood mixer achieves uniform spatial dispersion throughout large liquid volumes.
4. **Multi-Probe Optical Deconvolution**: Specific fluorescence signals were achieved across all four individual fluorophore probe channels (Probe A: HEX, Probe B: FAM, Probe C: Texas Red, and Probe D: Cy5). Furthermore, the 4-plex multiplex reaction produced positive amplification across all channels simultaneously (C_q = 11.95–15.49), validating multi-signature forensic verification.
5. **Commercial Product Integration & Non-Destructive Recovery**: Application of the taggant onto commercial packaging materials (pharmaceutical syrup bottles, cartons, tablet blister foils) and liquid formulations demonstrated practical compatibility. Surface swabbing with sterile cotton swabs in TE buffer enabled non-destructive recovery and positive downstream qPCR detection (PR samples, C_q = 3.99–5.23).

5.2 Conclusion

This project phase established and validated the complete laboratory pipeline for the **Indigenous DNA Tagging System for Industries**. From bacterial culture and enzymatic synthesis to 1.0-Liter bulk formulation and 4-channel multiplex qPCR verification, the system demonstrated:

- High amplification reproducibility and specificity at a 200 bp molecular target size.
- Complete spatial homogeneity across industrial carrier volumes.
- Covert, non-destructive forensic detectability from physical commercial packaging.
- Definitive multi-dye authentication capable of securing industrial supply chains against counterfeiting and unauthorized diversion.

The experimental protocols, recipes, and baseline verification data established herein provide a rigorous technical foundation for advancing the technology toward pilot-scale industrial deployment.